

The Shop Card

books-99 — what I built, and the mistake it prevents

You said you would get the low-limit card and stop using the master till for petty cash. That is the right call, and the card feature is now built. What follows is what it does, and the one thing I found while building it that would have cost you real money.

What happens now when Alyssa buys supplies

She takes the card off the hook, drives to the store, buys \$15 of bags, and comes back. She does not take cash, she does not leave a receipt in the till, and nobody has to remember anything. The charge appears in the bank feed and the books record it:

DEBIT	Store supplies	\$15.00	
CREDIT	Credit Cards Payable	\$15.00	

The master till stays at exactly \$1,000, all day, every day. That was the whole point of moving off petty cash, and it is now true by construction rather than by everyone being careful.

Notice the second line. Buying on the card does not spend cash — it creates a debt. The \$15 sits in Credit Cards Payable until you pay the bill.

The mistake I found, and what it would have cost you

When the statement gets paid from checking, your bank reports it as money going out. Before this slice, nothing in the system could tell that apart from an ordinary purchase. It would have read the description — "CHASE CARD PMT" — classified it as an expense like any other, and recorded this:

DEBIT	Some expense	\$15.00	
CREDIT	Checking	\$15.00	

That entry balances. The bank reconciles to the penny. And it is wrong, because you would have deducted that \$15 twice: once when Alyssa swiped, and again when you paid the bill. Every single card purchase, double counted. Meanwhile Credit Cards Payable would climb forever, because nothing was ever paying it down on the books — a card balance that only went up, for a card you were actually paying off every month.

I want to be direct about why this matters more for you than for most people. Under 280E your deductible expenses are already the part an examiner looks at hardest. An expense total inflated by a duplicated card bill is exactly the kind of finding that turns a look into an adjustment. This is filed as **D-78**.

The correct entry for paying the bill touches no expense account at all:

DEBIT	Credit Cards Payable	\$15.00	(you owe less)
CREDIT	Checking	\$15.00	(you have less)

Both sides are balance sheet. The expense was already recorded weeks ago, at the swipe. That is the entire idea of a card: the expense and the cash payment happen at different times, and the liability carries the gap between them.

How the system knows, and the trap inside the fix

It does not read the description. Bank descriptors change, and "CHASE" could just as easily be something you genuinely bought. Instead it uses the category your bank itself attaches to the row through Plaid — the issuer's own structured statement of what that transaction is. I pulled Plaid's published category list directly rather than assume the name.

That list is also where I found the trap. Their documentation says a card payment shows up as *"positive amounts for credit card subtypes and negative for depository subtypes."* In plain terms: **the same payment appears twice**, once on the checking feed and once on the card feed, pointing opposite ways. A naive fix would have booked both and paid the card down twice — the same double count, walking back in through the other door.

So exactly one copy counts. The books use the checking side, because that is where the real cash moves and that is the date your bank statement shows. The card's copy is recognised and deliberately skipped, and if you ever look at it the screen says so in words: *nothing was recorded here, and nothing is missing*.

One thing I was careful **not** to do: interest and finance charges are still recorded as real expenses. A rule greedy enough to treat those as transfers would quietly delete a legitimate deduction, and a missing deduction leaves no trace of itself. That boundary is pinned by its own test.

How hard I tried to break it

I wrote 20 deliberate sabotages of this feature and required the test suite to catch every one — reversing the entry, paying from the ATM account, letting the card bill reach the classifier again, swallowing the interest charge.

The first run caught 15 of 20. **Five got through, and those five were the valuable part.** Three were the same shape: the logic was perfectly correct and simply never reached, because the service was not reading the category column out of the database. Correct code that nothing calls looks identical to working code from every screen — and it would have left D-78 fully in place. I closed all five. It now catches 20 of 20.

Full suite: **12,885 tests across 504 files, all passing**, along with the type check, the pure self-tests, the citation check, the lint, and the ledger census. No database change was needed — your bank feed has been storing that category column since books-88.

One practical note

Put the card on the same Plaid connection as the operating account when you set it up. The books need to see both feeds to recognise both halves of a payment. If only the card is connected, the checking-side row never arrives and the liability will not come down.

When the card arrives, tell me the issuer and the limit and I will confirm the role is set correctly on the Plaid screen — that role is what routes purchases to Credit Cards Payable, and it is the one setting a person has to get right.